Lab Manual - LAB 4

Semester V – Fall

Group A and B

Course Title: Computer Networks

Bachelor of Science in Software Engineering

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LAB 4: IPv4 Address Configuration and Connectivity in Cisco Packet Tracer

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IPv4 Addressing and Basic Network Configuration Using Cisco Packet Tracer

Learning Outcomes

After this lab, you will be able to:

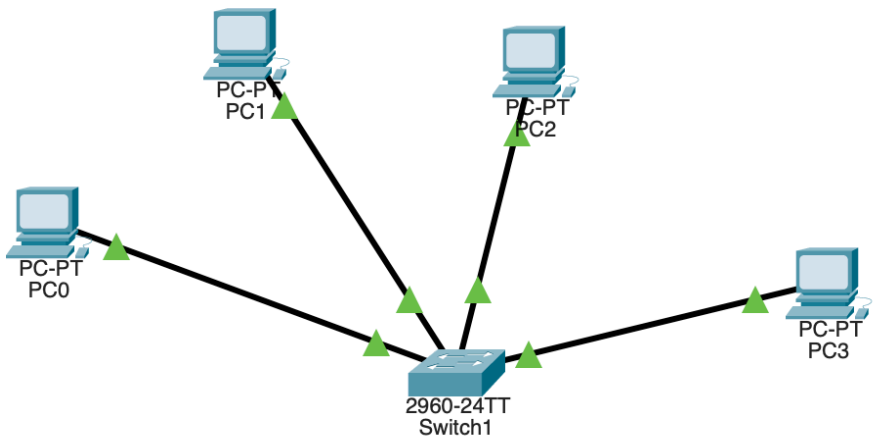
- Create a network containing two different IP networks.
- Configure IPv4 addresses on PCs.
- Configure router interfaces.
- Activate router interfaces using no shutdown.
- Configure default gateways.
- Test connectivity using ping.
- Verify router interfaces using show ip interface brief.

Task 1: Multiple PCs in the Same Network

A small computer lab has **four PCs connected to one switch**. All four computers belong to the same IP network.

- Build the topology.
- Assign IPv4 addresses.
- Configure the subnet mask.
- Determine whether all PCs belong to the same network.
- Test connectivity between the PCs.

Topology



IP Addressing

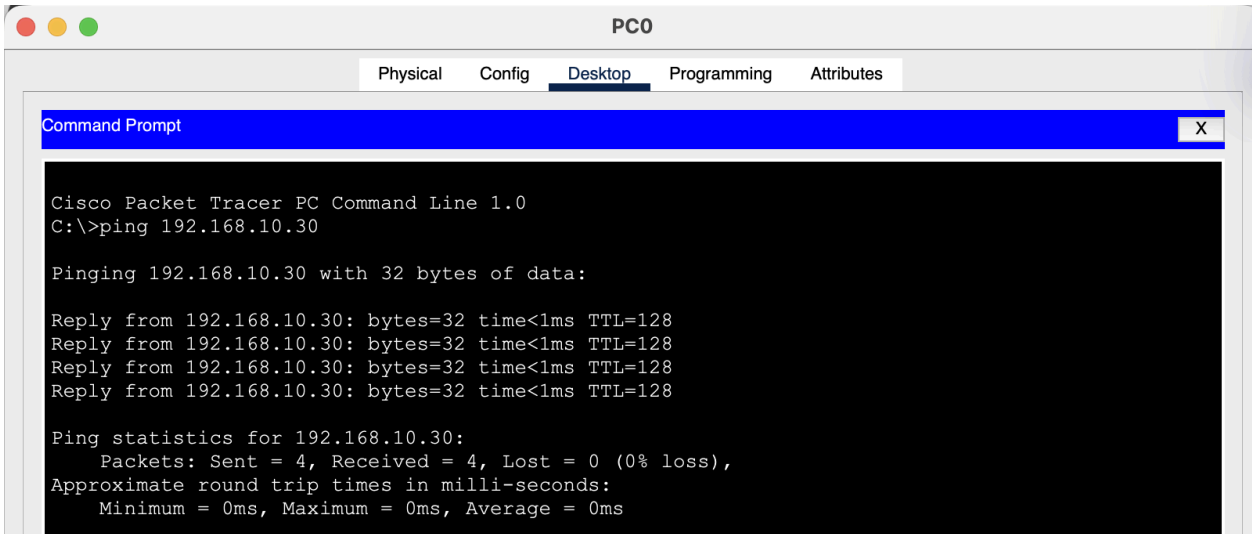
Device	IP Address	Subnet Mask
PC-0	192.168.10.10	255.255.255.0
PC-1	192.168.10.20	255.255.255.0
PC-2	192.168.10.30	255.255.255.0
PC-3	192.168.10.40	255.255.255.0

Tasks

1. Add **4 PCs** and **1 switch**.
2. Connect all PCs to the switch.
3. Configure the IP addresses.
4. Configure the subnet mask.
5. From PC-0, ping PC-1.
6. From PC-0, ping PC-2.
7. From PC-0, ping PC-3.
8. Record the results.

Expected Result

All four PCs should successfully communicate because they belong to **192.168.10.0/24.**



```
C:\>ping 192.168.10.40

Pinging 192.168.10.40 with 32 bytes of data:

Reply from 192.168.10.40: bytes=32 time<1ms TTL=128
Reply from 192.168.10.40: bytes=32 time<1ms TTL=128
Reply from 192.168.10.40: bytes=32 time<1ms TTL=128
Reply from 192.168.10.40: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

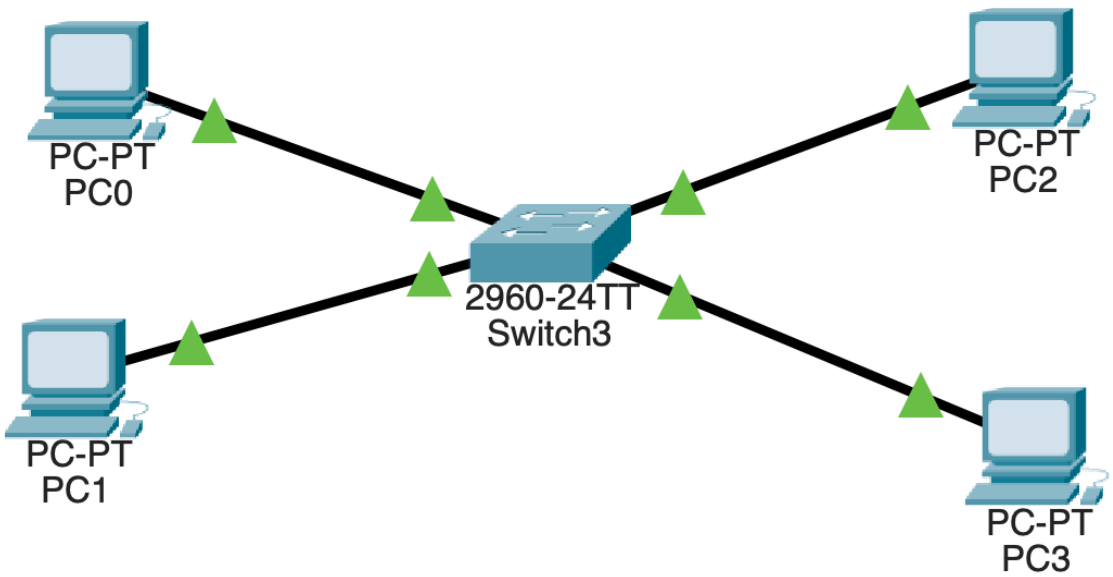
Task 2: Same Switch, Different Networks

A switch connects computers belonging to **two different departments**:

- Department A → Network 192.168.10.0/24
- Department B → Network 192.168.20.0/24

There is **no router** in this topology.

Topology



IP Addressing

Device	IP Address	Subnet Mask	Network
PC-0	192.168.10.10	255.255.255.0	192.168.10.0/24
PC-1	192.168.10.20	255.255.255.0	192.168.10.0/24
PC-2	192.168.20.10	255.255.255.0	192.168.20.0/24
PC-3	192.168.20.20	255.255.255.0	192.168.20.0/24

Before Testing

Test	Same Network?	Expected Result
PC-0 → PC-1	Yes	Successful
PC-2 → PC-3	Yes	Successful
PC-0 → PC-2	No	Failed
PC-1 → PC-3	No	Failed

Tasks

1. Build the topology.
2. Configure all four PCs.
3. Determine the network of each PC.
4. Predict which PCs should communicate.
5. Use ping to test your predictions.
6. Compare your predictions with the actual results.

Important Observation

Even though all four PCs are connected to the same physical switch, they belong to two different IP networks. The switch alone cannot provide communication between these different networks.

Task 03: Connecting Two Different Networks Using a Router

In this task, you will connect two different IP networks using a router. You will configure IPv4 addresses on PCs and router interfaces, assign default gateways, and test communication between the networks.

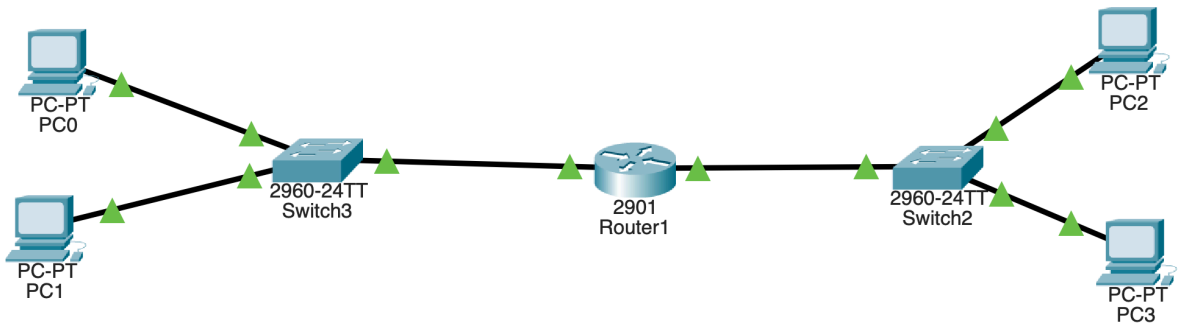
Consider a university with two computer labs.

- **Lab A** uses the network 192.168.10.0/24.
- **Lab B** uses the network 192.168.20.0/24.
- A router is required to allow communication between the two networks.

The topology will contain:

- 4 PCs
- 2 switches
- 1 router

Topology



IP Addressing

We will use two separate /24 networks.

Network A: 192.168.10.0/24

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.20	255.255.255.0	192.168.10.1

Router G0/0	192.168.10.1	255.255.255.0	—

Network B: 192.168.20.0/24

Device	IP Address	Subnet Mask	Default Gateway
PC2	192.168.20.10	255.255.255.0	192.168.20.1
PC3	192.168.20.20	255.255.255.0	192.168.20.1
Router G0/1	192.168.20.1	255.255.255.0	—

Important Note

After connecting the router to the switches, the router links may appear **red**. This is normal. Router interfaces are initially disabled. We will activate them during router configuration.

Configure Router

Now we will configure the router so that it can communicate with both networks.

1. Click: **Router1 → CLI.** You will see the router command prompt.
2. Enter Privileged Mode by typing **enable**. The prompt changes from **Router>** to **Router#**

```
Router>enable
Router#
```

3. Then enter configuration mode by typing **configure terminal.** The prompt changes to Router(config)#

```
Router#configure terminal
```

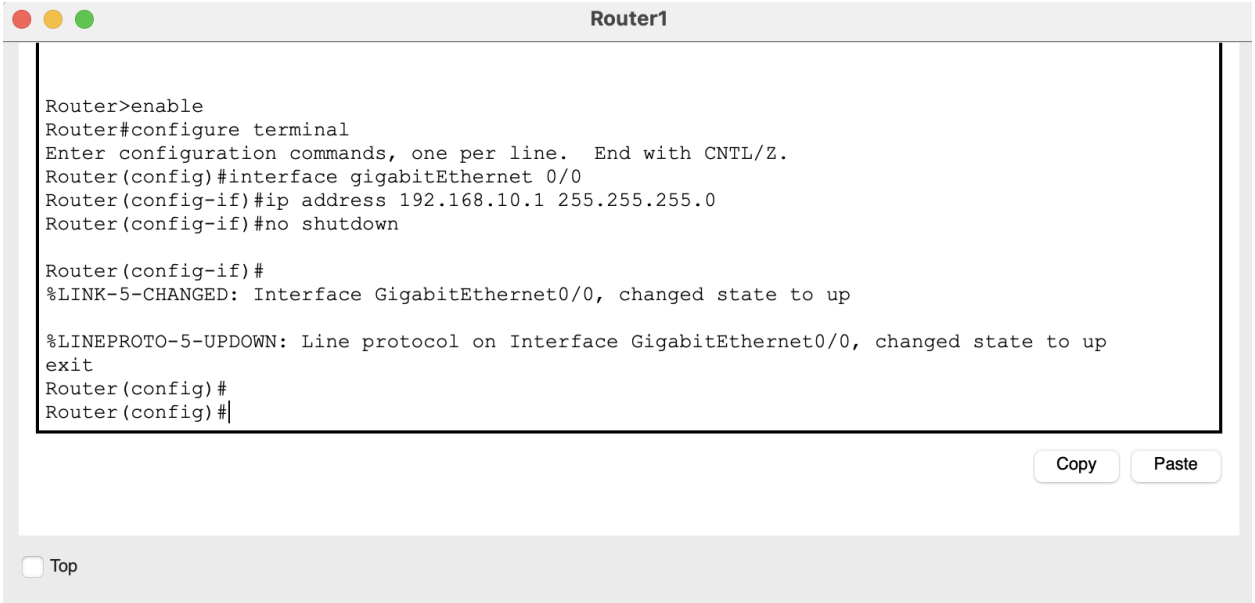
4. The first interface will connect Router1 to **Network A**.

```
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
```

5. **Why no shutdown?**

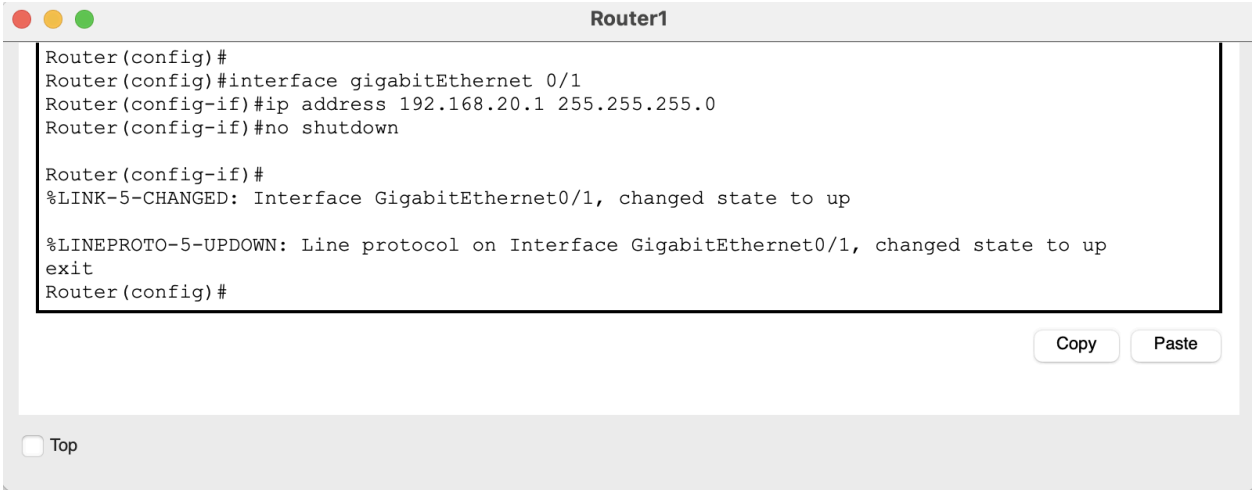
   A router interface can be administratively disabled. The command **"no shutdown"** activates the interface and will display a message indicating that the interface and line protocol have changed to **up**.

6. Then type exit.



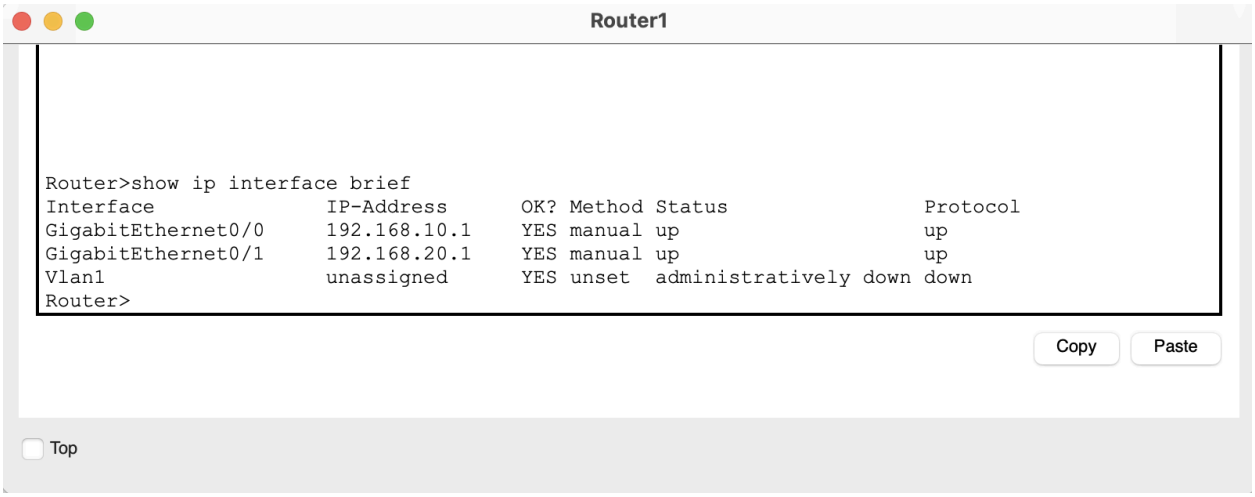
7. The second interface will connect Router1 to **Network B**.

```
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
```



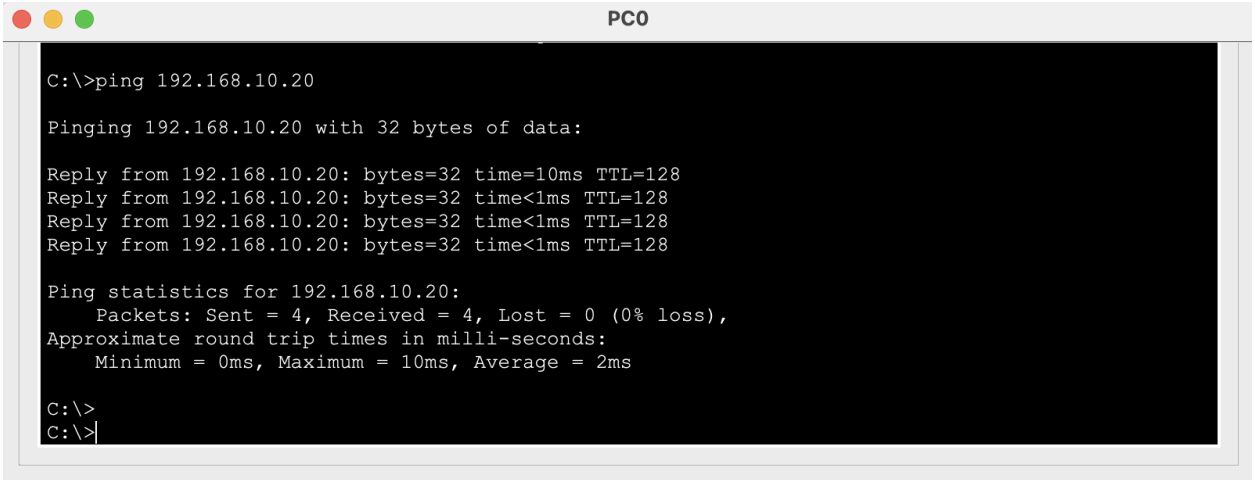
8. Verify whether both router interfaces are working:

```
Router>show ip interface brief
```

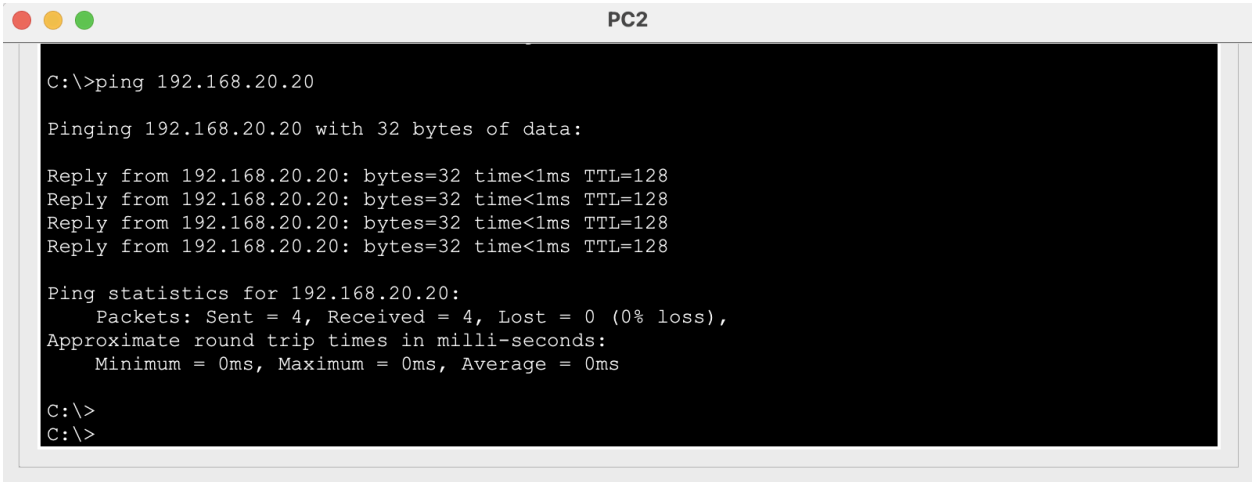


Test Communication Within Network A

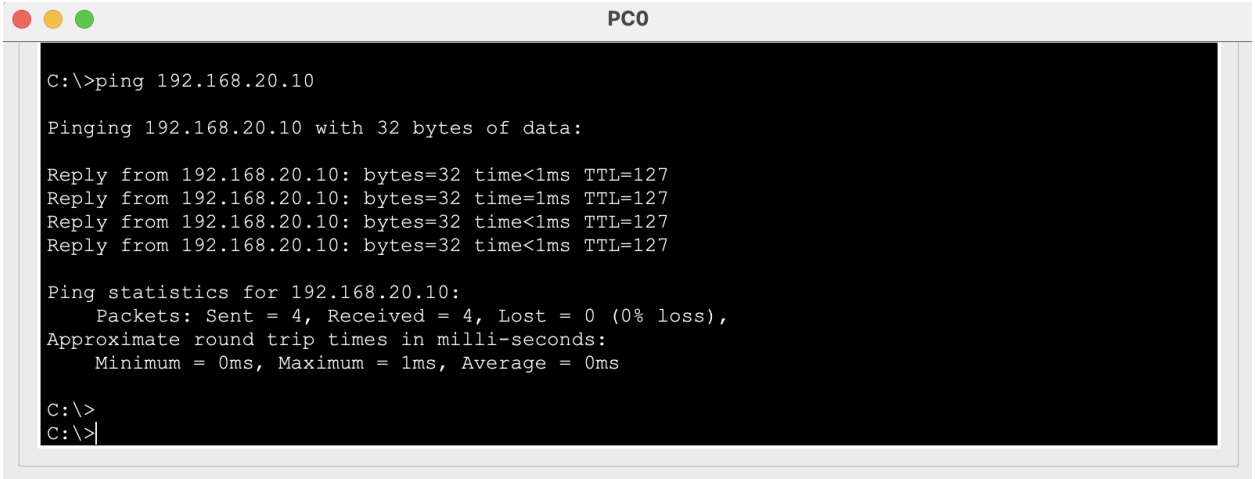
Before testing communication between different networks, first test devices within the same network. From PC-0, ping PC-1 (192.168.10.20).



Now Test Communication Within Network B. From PC2, ping PC-3 (192.168.20.20)



Now Test Communication Between Different Networks. From **PC0**, ping PC2 (192.168.20.10):



The path is PC0→Switch3→Router1→Switch2→PC2

Task 04: Troubleshooting an Incorrect Configuration

In this task, you will troubleshoot a network configuration problem. Use the same topology from Task 03. This time, PC2 has been given an **incorrect default gateway**.

Configure PC2 as follows:

Setting	Value
IP Address	192.168.20.10
Subnet Mask	255.255.255.0
Default Gateway	**192.168.10.1**

Do not change the configuration of the other devices.

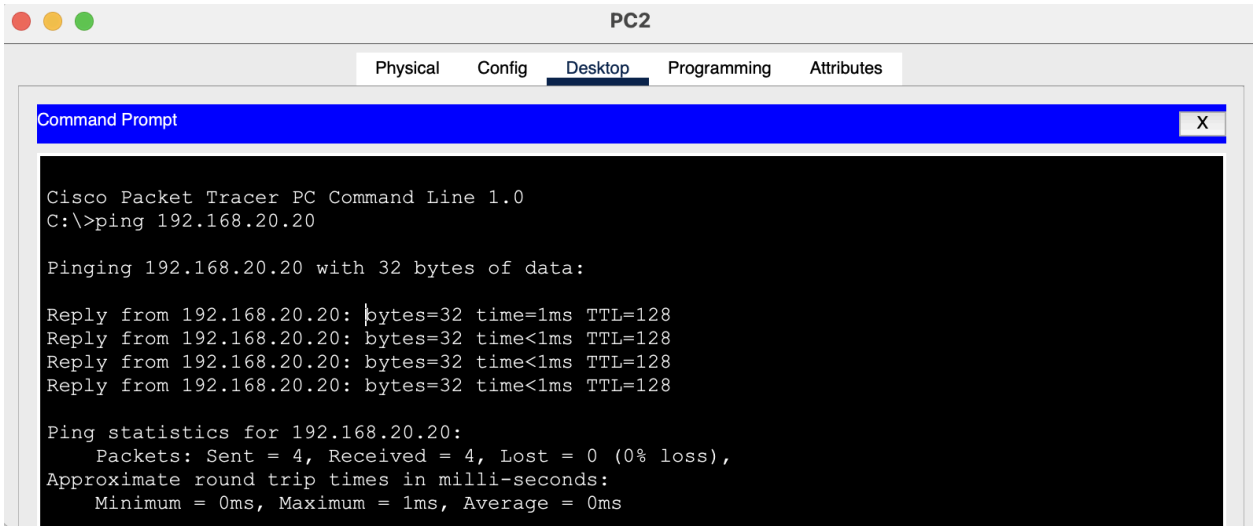
Identify the Problem

Look at PC2's configuration.

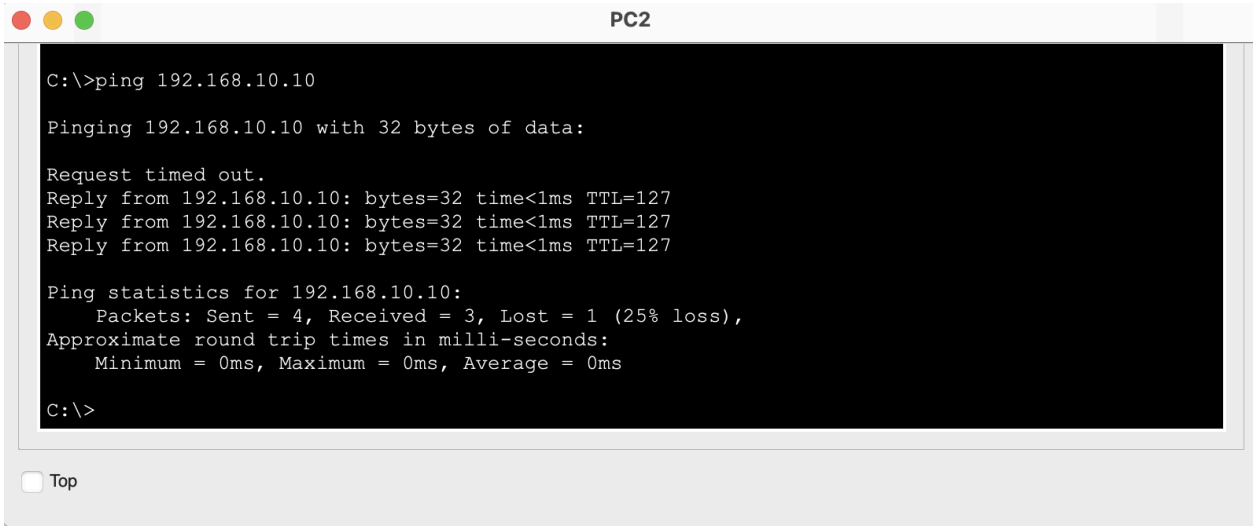
- PC2 has IP Address: 192.168.20.10
- Therefore, PC2 belongs to 192.168.20.0/24.
- However, its default gateway is: 192.168.10.1
- The gateway is on a **different network**.

Test the Incorrect Configuration

1. From PC2, test: ping 192.168.20.20. This should work because PC2 and PC3 are on the same network.

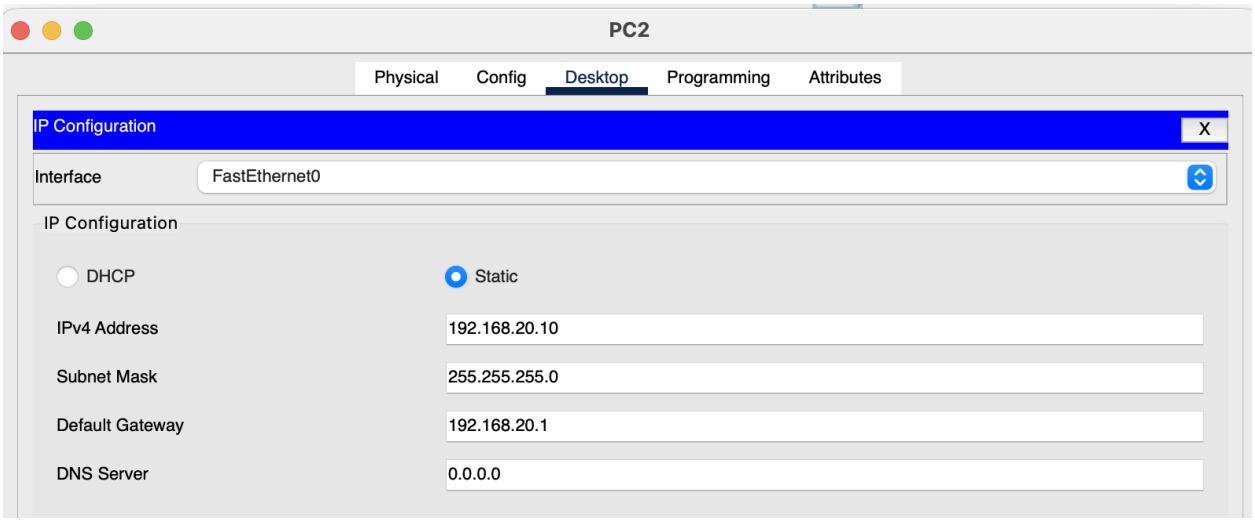


- Now test PC0. ping 192.168.10.10. The communication should fail because PC2 has an incorrect default gateway.

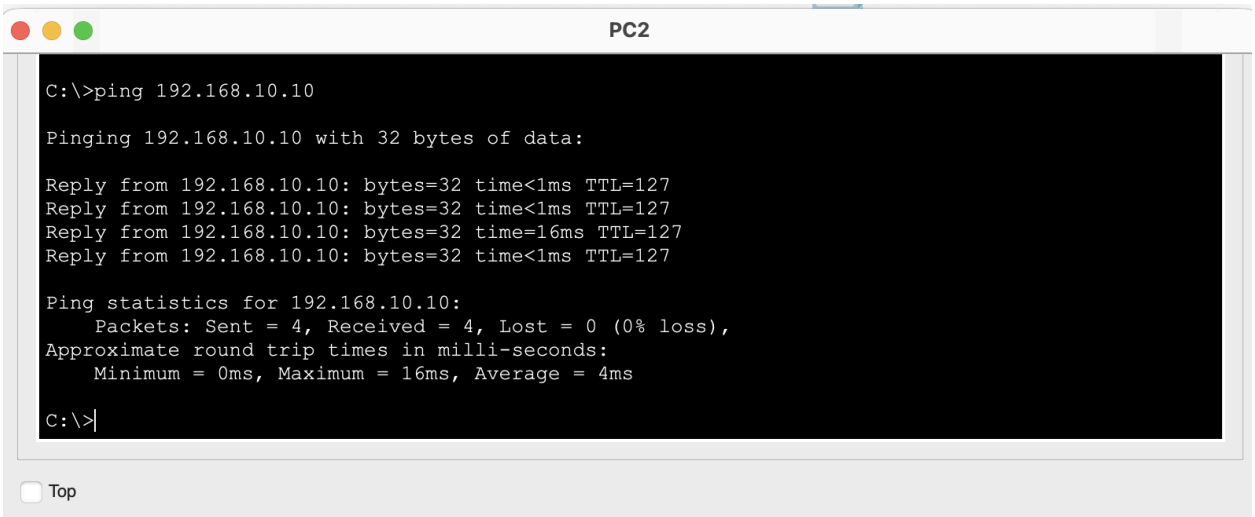


Correct the configuration and test again

- Change the Default Gateway of PC2 from 192.168.10.1 to 192.168.20.1. Do not change the IP address or subnet mask.



- Test the Corrected Configuration. From PC2, ping PC0 (192.168.10.10)



Commands

Command	Purpose
enable	Enter privileged mode
configure terminal	Enter configuration mode
interface gigabitEthernet 0/0	Select an interface
ip address ...	Assign an IP address
no shutdown	Activate an interface
show ip interface brief	Check interface status
ping <IP>	Test connectivity
exit	Exit the current mode
end	Return to privileged mode